BIWEEKLY INTERN PROGRESS REPORT

Document Intelligence & Retrieval Pipeline

Parsing benchmarks → evaluation harness → chunking strategy selection

Weeks 3–4 • Document Understanding Track

OVERVIEW

What this report covers

01

Week 3 Recap

Layout detection, OCR, and table-extraction benchmarks on synthetic catalog data

02

Week 4: Evaluation Harness

Built a hybrid-retrieval eval pipeline and benchmarked 5 chunking strategies

03

Week 4: Real-World Debugging

Fixed three ingestion / retrieval bugs on an unseen medical catalogue

04

Week 5 Plan

Benchmark candidate embedding models with the pipeline held constant

Parsing benchmarks: done and dusted

Layout detection, OCR, and table extraction were benchmarked on synthetic catalog data with zero-error ground truth. Clear winners emerged on every stage — these are now our go-to models for the pipeline.

✓ **Chapter closed — all three benchmarking tracks complete, validated, and locked in**

LAYOUT DETECTION

DocLayoutYOLO

GO-TO MODEL

Nemotron-Parse is only marginally more accurate, but DocLayoutYOLO is dramatically lighter (~15–30M vs. ~885M params) and faster — CPU-capable and the clear speed/accuracy winner.

TEXT / OCR

Tesseract

GO-TO MODEL

Dominant exact-match rate — most reliable for fully-correct field extraction.

TABLE EXTRACTION

Docling

GO-TO MODEL

Near-perfect structural recovery — the standout performer by a wide margin.

From parsing to retrieval: the evaluation harness

run_chunking_eval.py / local_judge_pipeline.py

Q

Ground-Truth QA Bank

54 golden Q&A pairs tagged with difficulty and reasoning type

R

Hybrid Retrieval

NumPy dense cosine + sparse lexical, fused via RRF (k=60) on local BAAI/bge-m3

C

Completer

Answers generated with llama-3.1-8b-instant on Groq

J

Independent Local Judge

qwen/qwen3-32b scores Ragas + DeepEval metrics — avoids generator self-preferencing bias

Engineering the harness for speed and scale

7 min → 20 ms

Embedding Caching

Pickled BGE-M3 embeddings cut repeat lookups from 7 minutes to 20ms

No re-generation

Completion Database

Cached generator outputs to avoid re-running Groq completions on repeat evals

4,800 tokens

Token-Aware Truncation

tiktoken-based prompt truncation capped at 4,800 tokens per context window

Reduced padding

Length-Sorted Batching

Batches grouped by sequence length to minimize padding waste during scoring

5 chunking strategies, scored across 54 Q&A pairs

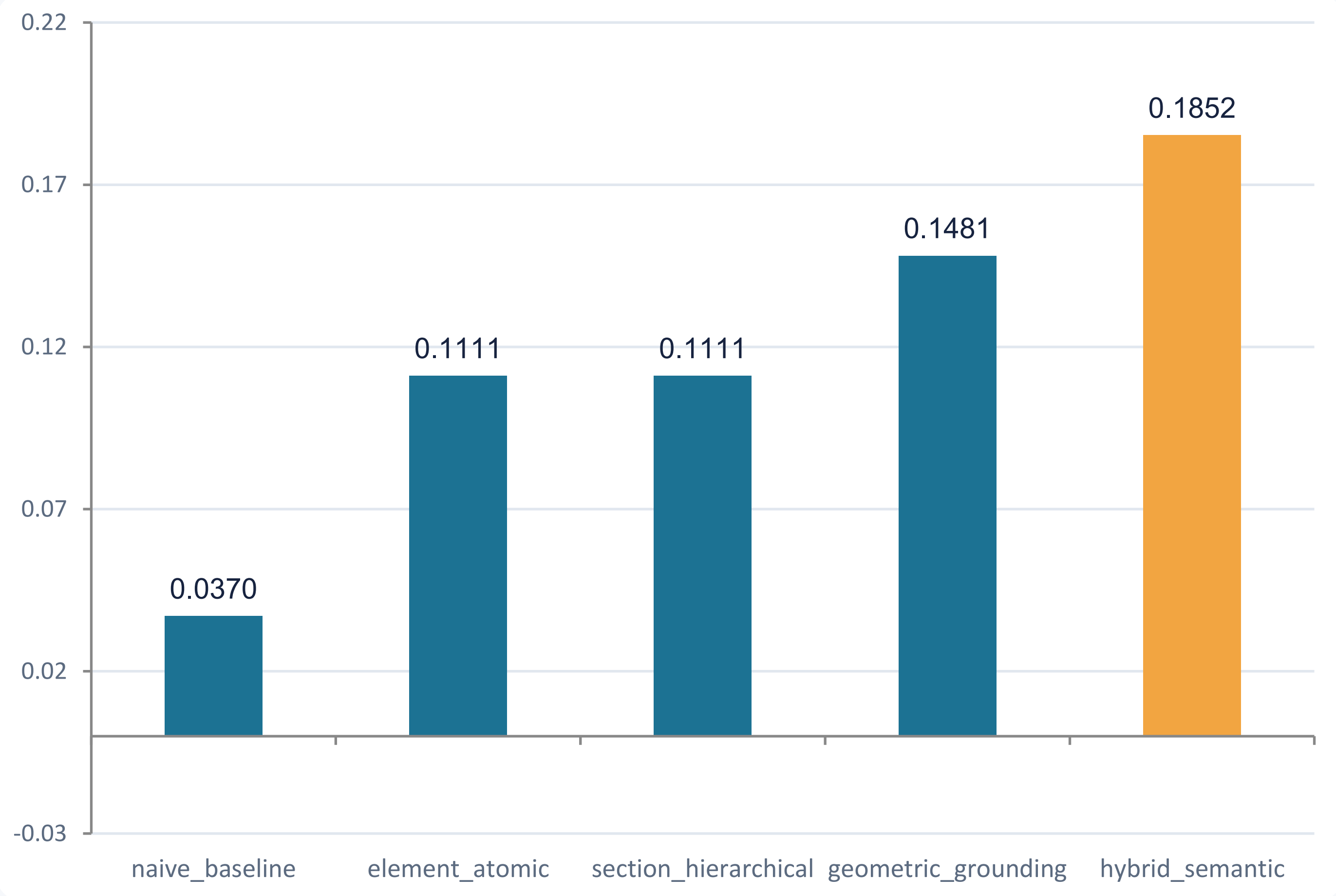
All strategies are open-source and implemented locally in-repo. Averages shown below (Ragas & DeepEval).

Strategy	Ragas Faithfulness	Ragas Ans. Relevancy	Ragas Ctx. Precision	Ragas Ctx. Recall	DeepEval Faithfulness	DeepEval Ans. Relevancy	DeepEval Ctx. Precision	DeepEval Ctx. Recall
naive_baseline	0.0370	0.0556	0.0370	0.0370	0.0370	0.0556	0.0370	0.0370
element_atomic	0.1111	0.1111	0.0889	0.0926	0.0926	0.1111	0.1074	0.1111
section_hierarchical	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111	0.1111
geometric_grounding	0.1481	0.1852	0.1444	0.1481	0.1481	0.1852	0.1444	0.1481
hybrid_semantic	**0.1852**	**0.1852**	**0.1450**	**0.1420**	**0.1852**	**0.1852**	**0.1635**	**0.1606**

hybrid_semantic wins on Faithfulness (0.1852) — 5x the naive baseline (0.0370)

Semantic-based boundary grouping preserves text coherence across chunk edges, consistently outperforming naive fixed-size splitting on every metric.

Faithfulness by strategy — the winning approach



STRATEGY SOURCE FILES

`chunking/strategies/`

- `naive_baseline.py`
- `element_atomic.py`
- `section_hierarchical.py`
- `geometric_grounding.py`
- **`hybrid_semantic.py`**

All five strategies are implemented locally in the repository — no external dependencies.

Stress-testing on an unseen 50-page medical catalogue

Medical_004_full.pdf — pages 50–100, held out from training/tuning

1

Page-Shift Ingestion

A 1-page shift in Docling's table parsing was fixed by re-extracting tables against corrected page grids.

2

Grounded VLM Captions

Stale figure captions on pages 24–25 were corrected by re-captioning with Llama-4-Scout, grounded in the updated tables.

3

Contextual Retrieval Disconnection

Side-by-side tables lost their headings. Added a y-axis proximity alignment algorithm linking tables to the closest preceding title / figure-caption.

Result: fully correct RAG answers on target queries

100%

correct RAG answers on target queries

Example: Crile Hemostats

After the y-axis proximity fix, the pipeline correctly linked the product title to its specification table — even in a side-by-side column layout.

1-1/4"

jaw length

5-3/4"

overall length

Both specs retrieved correctly and attributed to the right product after the ingestion fixes above.

WEEK 5 PLAN

Next: benchmarking candidate embedding models

Pipeline held constant — hybrid_semantic chunker, Groq generator, local Qwen judge — embedding model is the sole variable.

1. **BGE-M3 (Baseline)**
2. Qwen3-Embedding-8B
3. NV-Embed-v2
4. Gemini Embedding
5. Cohere embed-v4
6. Voyage-3-large
7. voyage-4
8. nomic-embed-text
9. Granite-vision embedding
10. Qwen3-VL

WEEK 5 PLAN

Benchmarking scope

R

Retrieval Modes

Dense retrieval vs. sparse keyword weighting vs. hybrid RRF fusion — including a new BM25 indexing layer for models without native sparse output.

Q

Quality Metrics

Hit Rate and Mean Reciprocal Rank (MRR) tracked against the same 54-question golden bank used this week.

S

System Footprint

Latency, throughput, and memory measured per model to weigh accuracy gains against deployment cost.

THANK YOU

Questions & Discussion

Next check-in: Week 5 — embedding model benchmark results